Group members:
1. SABRINA HENG WEI QI
2. WOO CHENG SHUAN

SECJ2013 - DSA

2024/2025-1

QUIZ 1

Given that two functions (**funA** and **funB**) have been defined as follows:

```
void funA(int n) {
  cout << "funA - " << n << "\n";
  n = n + 1;
  if (n < 5) {
    if (n % 2 == 0) {
      funB(n);
    } else {
      funA(n);
    }
  }
}
```

```
void funB(int n) {
  cout << "funB - " << n << "\n";
  n = n + 1;
  if (n < 5) {
    if (n % 2 == 0) {
      funB(n);
    } else {
      funA(n);
    }
  }
}
```

Study the funA and funB functions above. Identify the followings:

1. Instruction to check if the terminal case has been reached: ________if(n<5)_______________________
2. Instruction to change the size of the problem so the terminal case can be reached: ______n = n+1____________

Trace the recursive calls of the funA and funB functions inside the following main function of a C++ program:

```cpp
int main() {
  funA(1);
  return 0;
}
```

funA(int 1) ;

1

Output:
funA - 1

```
void funA(int 1){
   cout << “funA –” << 1 << “\n”;
   n = 1+1;
   if (2<5){
      if (2%2 == 0){
         funB(2);
      } else {
         funA(2);
      }
   }
}
```

2

Output:
funA – 1
funB - 2

```
void funB(int 2){
   cout << “funB –” << 2 << “\n”;
   n = 2+1;
   if (3<5){
      if (3%2 == 0){
         funB(3);
      } else {
         funA(3);
      }
   }
}
```

3

Output:
funA – 1
funB – 2
funA - 3

3

```
void funA(int 3){
    cout << “funA –” << 3 << “\n”;
    n = 3+1;
    if (4<5){
        if (4%2 == 0){
            funB(4);
        } else {
            funA(4);
        }
    }
}
```

Output:
funA – 1
funB – 2
funA – 3
funB - 4

4

```
void funB(int 4){
    cout << “funB –” << 4 << “\n”;
    n = 4+1;
    if (5<5){
        if (5%2 == 0){
            funB(5);
        } else {
            funA(5);
        }
    }
}
```

Reach the terminal case